

The Singh family is building a pool. They were approved by their bank for a loan of \$85,000 with a simple interest rate of 2.25%.

1. Write an equation that models the Singh's loan.

$$A = P(1 + rt)$$

$$A = 85,000(1 + 0.0225t)$$

2. Rewrite the equation in slope-intercept form.

$$A = 85,000 + 1912.5t$$

$$A = 1912.5t + 85,000$$

3. Complete the table.

Key Feature	Value	Interpretation in Context
y-intercept	(0,85000)	When the loan begins, the balance is \$85,000.
slope	1912.5	The loan balance increases by \$1912.50 per year.

4. What is the total the Singh's will have to pay if they pay off the loan in 5 years?

$$1912.5(5) + 85,000$$

$$9,562.5 + 85,000$$

$$\$94,562.50$$

5. How many years did it take the Singh's to pay off the loan if the total they paid was \$109,000?

$$\begin{array}{rcl} 109,000 & = & 1912.5t + 85,000 \\ -85,000 & & -85,000 \end{array}$$

$$\frac{24,000}{1912.5} = \frac{1912.5t}{1912.5}$$

$$13 \approx t$$

Sophie just graduated college and is starting her first job. She is investing money into a savings account to plan for retirement. She has \$10,000 to invest in a savings account with a 4.50% APR.

6. Write an equation that models Sophie's savings account.

$$A = 10,000(1 + 0.045t)$$

7. Write the equation in slope-intercept form.

$$A = 10,000 + 450t$$

$$A = 450t + 10,000$$

8. Complete the table.

Key Feature	Value	Interpretation in Context
y-intercept	(0,10000)	There is \$10,000 in Sophie's saving account before any interest is earned.
slope	450	Each year \$450 will be added to Sophie's account.

9. How much money will be in the savings account if Sophie wants to retire 30 years later?

$$450(30) + 10,000$$

$$13,500 + 10,000$$

$$\boxed{\$23,500}$$

10. How many years would Sophie have to work to retire with \$30,000?

$$\begin{array}{rcl} 30,000 & = & 450t + 10,000 \\ -10,000 & & -10,000 \end{array}$$

$$\frac{20,000}{450} = \frac{450t}{450}$$

$$\boxed{45 \text{ years} \approx t}$$